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ECOLOGY

Coral–Algal Competition on Damaged Reefs

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Abstract—Natural and anthropogenic catastrophes occurred at the end of the previous and in the beginning of the current centuries at the coral reefs of the World Ocean, and their consequences for the tropical shelf ecosystems have been described based on published data and our own investigations. It has been shown that in recent decades coral populations on reefs of tropical and subtropical regions of the World Ocean have been reduced by 80%, and in some areas have completely vanished. The biodiversity of reef ecosystems has been considerably reduced. The main reason for such changes is a1–2°C increase in the temperature of surface waters in comparison with the monthly mean temperature in the hot season. The fate of the damaged coral reefs is under discussion. It is thought that in clean waters partially damaged coral reefs can recover, whereas in waters polluted as the result of human activity they collapse. The rate of coral reef restoration depends on the hydrological and hydrochemical conditions, frequency of natural calamities and competitive interrelation of algae and corals on the damaged sites of coral reefs. The nature of competitive interrelation between algae and corals is considered, viz., the dynamics of obliteration of damaged and dead coral colonies by various algal species, mechanisms of competitive interrelation, effects of the environment on the competitive ability of corals and algae, the internal and external conditions for victory in competitive activity. It has been suggested that coral reefs can be restored through temporary transformation into a vegetable reef. In the absence of natural calamities damaged reefs can be clearly restored to their original or altered state over several decades, but only in clean waters.

Key words: algae, corals, competition, natural and anthropogenic catastrophes.
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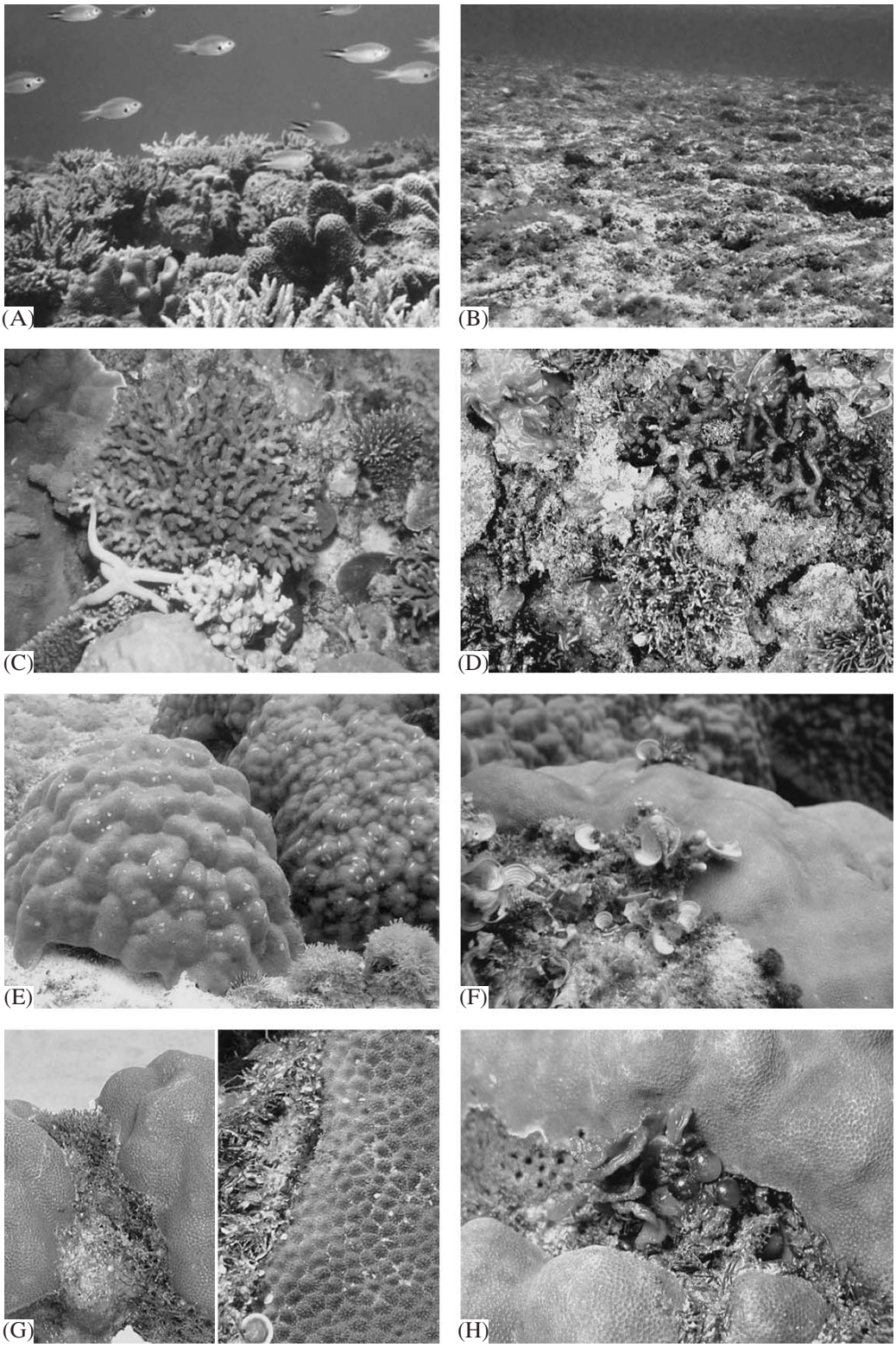
There are several reviews devoted to the description of algae and coral interrelation on coral reefs. Thus, Miller [103] analyzes the effects of local and latitudinal external factors on competitive interrelations between sea algae and corals and on the composition and structure of reef ecosystem communities. McCook [93] described the competitive interrelation of algae and corals on adverse and degenerating coral reefs of Australia. In a paper by McCook [95] the mechanisms of competition of algae and corals for space, substrate, nutrient resources, and for the conditions needed for successful competitive activities of algae with corals and vice versa were discussed.

In our review we discuss some data on the competitive interrelation of algae and corals on reefs that have been damaged as the result of natural and anthropogenic catastrophes, from the first days of the damage to their restoration or complete degradation.

Natural and Anthropogenic Catastrophes on Coral Reefs

During the last decades of the previous century and in the beginning of the current one coral reefs of the World Ocean have undergone catastrophic changes under the influences of natural and anthropogenic factors [21, 23, 50, 52, 61, 110, 119, 151, 152, 157, 158]. In the shallow areas of tropical and subtropical zones of the World Ocean at the present time up to 80% of hermatypic coral populations have been lost, and together with them other animals, which are coral symbionts or coral reef associates [38, 39, 67, 79] (figure, A and B). The main cause of coral reef loss is the 1–2°C water temperature increase in the tropical and subtropical zones of the World Ocean in comparison with the monthly mean temperature within the hot season [61, 81, 89], which is probably caused by such global climatic changes as warming [21, 120, 156], and the El Nino effect [37, 53, 54, 127]. Coral loss upon a water temperature increase occurs due to disruption of the symbiotic relation between animals-polyps and their endosymbiotic algae-zooxanthellae, resulting in coral bleaching and degradation of host cells, or in being extruded [27-29, 106, 135, 142,]. Thus, corals exposed to elevated temperatures bleach partially or completely, and may be lost in the future. In some cases due to the long-term effects of high temperature (1–2 months) all coral can be lost in shallow waters [35, 88].

In 1998 the greatest coral bleaching of the century took place [125, 158], which affected almost all the coral reefs of the Indo-Pacific [24, 48, 110] (figure, A and B) and partially of the Atlantic [22, 54].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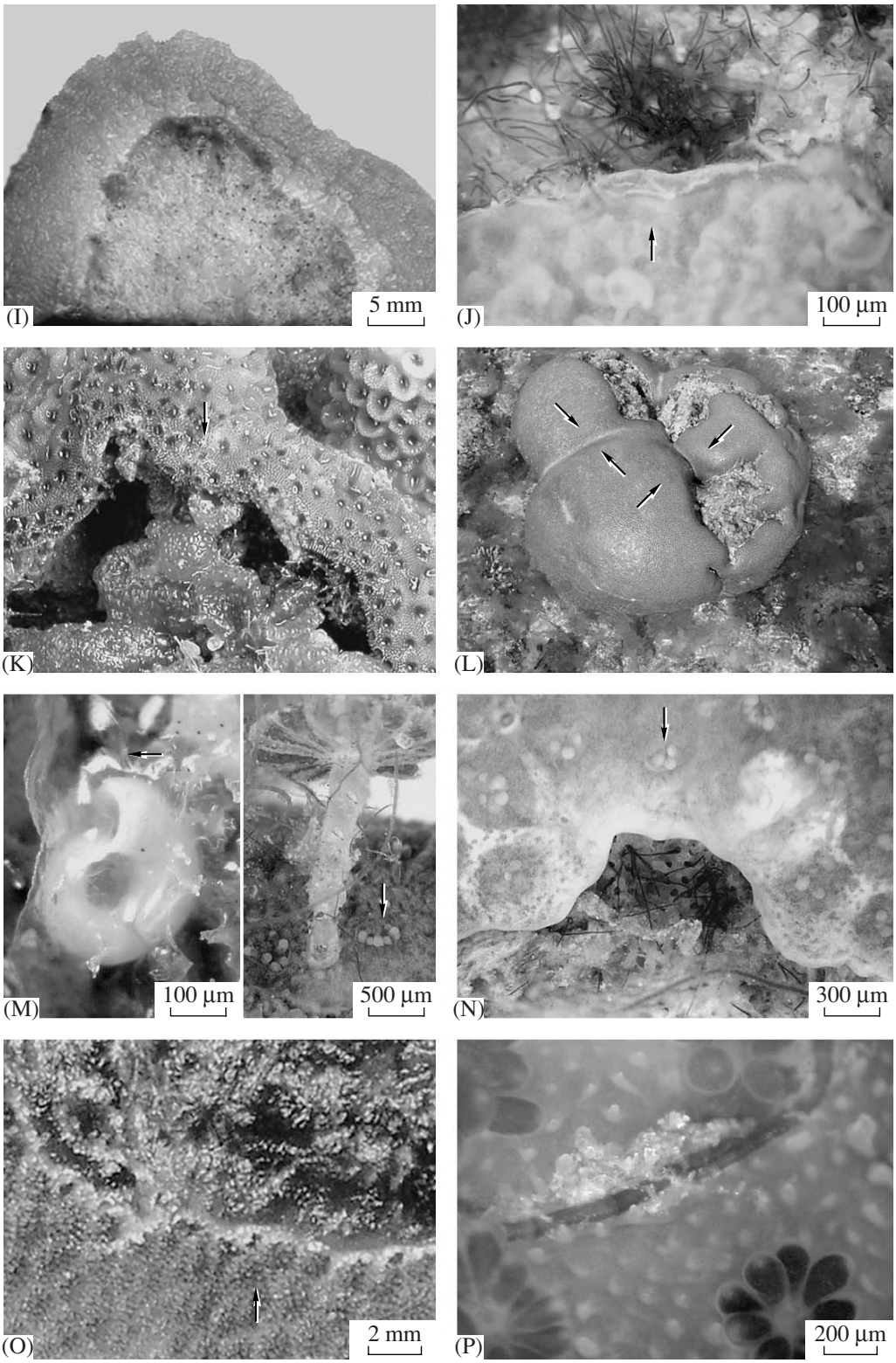
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5 mm
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(N)
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(O)
2 mm
(P)
200 μm

Figure. Corals and algae on the fringing coral reef of Sesoko Island. The arrows show the direction of the dead parts of the wound overgrown with coral tissue (polyps).

A, B—the fringing coral reef of Sesoko Island (southwestern coast, depth 2.5 m): A—July 1995; B—2003; C, D—reef flat of esoko Island opposite the Marine Biological Station (depth 1.5 m): C—1995; D—2003; E—colonies of the massive coral *Porites lutea* undamaged in 1998. White spots on the colony's surface are damage from coral-eating fish; F—a colony of the massive coral *P. lutea* damaged in 1998. The dead part of the colony is 100% overgrown with *Padina australis, Halicorine wrightii* and *Lobophora variegata* algae; G—damage on the upper surface of the *P. lutea* colonies overgrown with algal turf: at the left—algal turf with prevalence of the red alga *Jania adhaerens*, at the right—*Hypnea* spp. and *J. adhaerens*. Bleached polyps (these acquire a green color due to the presence of endolithic algae in corallites) at the border with a dense community of algal turf; H—a wound in the lower part of a *P. lutea* colony, overgrown with the green algae *Halimeda macrophysa* and *Bornetella sphaerica*; I—restoration of echanical damage on a *P. lutea* coral colony after one month of exposure in the sea, the first algal and animal settlers are not serious obstacles for coral tissue advance along the wound surface; J—*Porites cylindrical* coral overgrows the first settlers—filamentous algae. Transparent coral tissue, covering algae, can be clearly seen; K—incrusting coral *Acropora* sp., overgrowing a wound and algae (*Dicryosphaeria cavernosa, J. adhaerens*); L—massive coral *Goniastrea aspera*, damaged in 1998 (from the intertidal zone opposite the Marine Biological Station at Sesoko, the photo was made in 2003). The arrows show the direction of the colony's growth and wound areas, completely overgrown with polyps; M—types of overgrowth of the dead part of the wound with polyps on the *P. lutea* colony: at the left—coral tissue, not containing zooxanthellae yet and covering the substrate and a *Spirorbis* (Polychaeta) pipe, at the right—coral tissue, surrounding a stalk of the green alga *Parvocaulis parvula*; N—*Lyngbya majuscula* (Cyanobacteria) is an obstacle impeding polyp restoration on the dead part of the wound of *P. lutea* coral; O—*Lyngbya semiplena* is an insurmountable obstacle for advance of *P. cylindrica* coral tissue along the dead part of the wound; P—*Centroceras clavulatum* on the surface of *Acropora* sp. coral, formation of calcium carbonate crystals around the thallus indicates the engulfing of living algae in the coral skeleton.

Mass or partial loss of corals on reefs cannot be caused only by a water temperature increase. Powerful waves produced by strong storms and tsunamis can completely ruin reefs, and sometimes wash away small flat islands [46, 71, 72]. Cyclones, bringing abundant precipitation, desalinate and pollute shallow waters, especially in estuarine areas, which also results in coral bleaching and loss [36, 122, 143].

Heavy damage and even loss of coral colonies can be a consequence of a catastrophic increase of the population density of the sea-star *Acanthaster*, consuming living tissue of scleractinian corals [26, 161]. Predatory gastropods *Drupella* spp., *Habromula spinosa*, and several others also cause damage to colonies of hermatypic corals [153, 161]. Coral-eating fish shave living tissue of corals together with corallites (skeleton). Coral reefs are also lost in the areas of human activity, often as a result of water pollution with domestic and agricultural wastes [31, 47, 90, 103, 137, 155]. When building waterside structures, corals can be damaged with sand and mud dredge, silted with river drifts, especially in the basins of deforestation; corals also suffer from mechanical damage from ships and anchors, and they are broken by diving tourists [30, 63, 64, 85, 126]. At present it is difficult or probably impossible to find a sound intact coral reef [58].

Colonization of Dead and Damaged Corals by Algae

Dead coral colonies and wounds on damaged colonies are rapidly populated with algae, sponges, hydroids and other organisms (figure, A–H). As can be seen from field observations and experiments conducted in the sea and aquaria [43–45, 146, 147], microscopic diatoms, microscopic and thin-filamentous blue-green and green algae (figure, I-P), as well as tunicates, foraminifers, and small polychaetes (figure, I-M, table) are primarily the first settlers, both on dead coral and on wounds of living colonies.

The possibility for colonization and its rate depends first of all on the extent of colony damage. Thus, after the mass coral bleaching on the Great Barrier Reef of Australia heavily damaged coral tissue was lost, and coral skeletons cleared of tissue were populated with algae. However, algae can colonize not only the dead coral skeleton, but also living tissue, weakened (bleached) by high temperatures [43], extremely severe shading [146] or by diseases [134]. Weakened coral tissue is unable to purify itself by means of pulsating movements of hairs on its surface, tentacles, or burial of algal spores and fragments in its skeleton. According to the observations of Diaz-Pulido and McCook [43], weakened and bleached coral tissue becomes overgrown first of all with green-blue algae of the *Spirulina* genus. According to our observations [146], coral from *Porites* genus weakened by long-term deficiency of light tissue was overgrown with the red filiform algae *Centroceras clavulatum* and *Ceramium* spp. At the same time, mechanically damaged coral tissue was not overgrown with algae, and algal thalli fragments and spores occurring on it turned out to be entombed in the colony skeleton [146].

On dead coral colonies and on dead wound areas germinating algal spores appeared as soon as during the first week after the damage, and on the 15–30th days after the damage from 15 to 30 algal species were found, which occupied 1–7% of the substrate area (see the Table) [147]. The first settlers on a free substrate were mainly diatoms and green-blue algae, and they formed there dense, compact communities (figure, I and J). Green, brown and red filiform algae also formed dense algal communities, resembling turf or mat [34, 146].

At the earliest stages of settlement of an algae the substrate has a definite importance [59]. Thus, in our

Species composition and abundance of algal settlers on mechanically damaged colonies of *Porites lutea* and on coral pellets during formation of algal communities under moderate shading (20–30% PAR) in the sea and aquaria

List of species	Morpho-functional forms	1st month, March		3rd month, May			6th month, August	55th month, March	60th month, May
		SE	AE	SE	AE	SpE	SpE	DbS	DbS
BACILLARIOPHYTA									
Fragillaria sp.	M	–	–	–	+	–	–	–	–
Asterionella japonica Cleve	M	–	–	–	++	–	+	–	–
Diatoma sp.	M	+	+	++	+	–	–	–	–
Navicula sp.	M	–	–	–	+	–	+	–	++
Nitzschia closterium (Ehrenberg) W. Smith	M	–	–	–	++	–	+	–	++
N. longissima (Brebisson) Ralfs	M	+	+++	++	+++	++	–	+	+
N. seriata Cleve	M	–	–	–	++	++	–	–	–
N. sigma	M	–	–	–	–	–	+	–	++
Licmophora abbreviata Cleve	M	+	+	++	+++	+	++	++	++
Tabellaria sp.	M	–	–	–	–	–	–	–	+++
CYANOBACTERIA									
Chroococcus turgidus (Kützing) Naegeli	M	–	–	–	–	–	+	–	–
Dermocarpa acervata (Setchell & Gardner) Pham Hoang Ho	M	–	–	+	+	–	++	–	–
Aphanocapsa littoralis Hansgirg	M	–	–	–	+	–	–	–	–
Lyngbya confervoides C. Agardh ex Gomont	FT	–	–	–	+	–	–	–	++
L. majuscula (Dillwyn) Harvey ex Gomont	FT	–	+	–	+++	–	–	–	+++
L. polychroa (Meneghini) Rabenhorst [= *Lyngbya sordida* (Zanardini) Gomont]	FT	–	–	++	–	++	–	+++	–
L. semiplena (C. Agardh) J. Agardh ex Gomont	FT	–	–	++	++	–	+	–	+
Lyngbya sp.	FT	–	+	–	+	++	–	+++	–
Oscillatoria margaritifera (Kützing) Gomont	MF	–	–	+	+	–	–	–	–
Oscillatoria sp.	MF	–	+	–	+++	+	++	++	++
Phormidium corium (C. Agardh) Gomont	MF	–	–	–	+	–	+	+	++
P. limosum (Dillwyn) D.C. Silva (=*Oscillatoria tenuis* Agardh)	MF	–	+	–	++	–	–	–	–
Phormidium sp.	MF	–	+	–	+	–	–	–	++
Leptolyngbya tenuis (Gomont) Anagnostidis & Komarek [=*Phormidium tenue* (Meneghini) Gomont]	MF	–	+	+	–	–	+	+	+
Spirulina major Kützing	MF	–	+	–	+	–	–	–	++
S. subsalsa Oersted	MF	–	–	–	+	–	+	–	+
S. subtilissima Kützing	MF	–	–	–	–	–	–	–	–
S. tenerrima Kützing	MF	–	–	–	+	–	+	–	–
Nostoc commune Vaucher	MF	–	–	–	–	–	–	+	–
Calothrix confervicola (Dillwyn) C. Agardh	MF	+	+	+	+	++	–	+	–
C. contarenii (Zanardini) Bornet & Flahault	MF	–	–	–	+	–	+	–	+
Calothrix sp.	MF	–	++	–	+++	++	++	–	++
Dichotchrix sp.	MF	+	+	+	++	–	++	–	+
Tolypothrix sp.	MF	–	++	+	+++	–	+	–	+
CHLOROPHYTA									
Micrasterias sp.	M	–	–	–	–	–	–	+++	–
Entocladia viridis Reinke	En	–	–	–	++	++	–	++	–
Pringsheimiella scutata (Reinke) Marchewianka	M	–	–	–	–	–	–	–	–
Monostroma nitidum Wittrock	Pl	–	–	+	+	–	–	–	–
Ulva clathrata (Roth) C. Agardh [=*Enteromorpha clathrata* (Roth) Greville]	FT	+	+	+	++	+	+	++	+
U. compressa Linnaeus [= *Enteromorpha compressa* (Linnaeus) Nees]	FT	+	+	+	++	+	+	++	+

Table. (Contd.)

List of species	Morpho-functional forms	1st month, March		3rd month, May			6th month, August	55th month, March	60th month, May
		SE	AE	SE	AE	SpE	SpE	DbS	DbS
Ulva sp.	Pl	–	–	+	+	–	+	+	+
Anadyomene wrightii Harvey ex J.E. Gray	MFl	–	–	–	–	–	–	–	+
Chaetomorpha linum (O.F. Müller) Kützing	F	+	+	–	–	–	–	–	–
Cladophora laetevirens (Dillwyn) Kützing	F	–	–	–	–	–	+	+	–
C. vagabunda (Linnaeus) Van den Hoek [=*C. fascicularis* Linnaeus]	F	+	+	++	–	+	+	–	–
Cladophora sp.	F	–	+	–	–	–	–	+	++
Rhizoclonium riparium (Roth) Harvey [=*R. implexum* (Dillwyn) Kützing]	F	–	–	–	+	–	+	+++	++
Boodlea composita (Harvey) F. Brand	S	–	–	–	+	–	–	–	+
Phyllodictyon anastomosans (Harvey) Kraft & M.J. Wynne [=*Struvea anastomosans* (Harvey) Piccone & Grunow ex Piccone]	MFl	–	–	–	–	–	–	–	+
Cladophoropsis membranacea (Hofman Bang ex C. Agardh) Børgesen	F	–	–	–	–	–	–	+	–
C. modonensis (Kützing) Børgesen	F	–	–	–	–	–	–	+	–
C. fasciculatus (Kjellman) Wille [=*C. sundanensis* Reinbold]	F	–	–	–	–	–	–	+	–
Cladophoropsis sp.	F	–	–	–	+	–	–	+	–
Dictyosphaeria cavernosa (Forsskål) Børgesen	B	–	–	–	–	–	–	–	–
Siphonocladus rigidus M. Howe	F	–	–	+	+	–	–	+	–
Ventricaria ventricosa (J. Agardh) Olsen & J. West	B	–	–	–	–	–	+	–	+
Bryopsis ryukyuensis Yamada	F	–	–	–	–	–	–	+	–
Derbesia fastigiata W.R. Taylor	FT	–	–	+	+	–	–	–	–
D. marina (Lyngbye) Solier	FT	–	++	+	++	–	–	–	–
Codium adhaerens C. Agardh	M	–	–	–	–	–	+	–	+
C. intricatum Okamura	M	–	–	–	–	–	–	–	+
C. repens P.L. Crouan & H.M. Crouan	M	–	–	–	–	–	–	–	+
Caulerpella ambigua (Okamura) Prud'homme van Reine & Lokhorst [*Caulerpa ambigua* Okamura; *Caulerpa vickersiae* Børgesen]	MF	–	–	–	–	–	–	++	–
Caulerpa nummularia Harvey ex J. Agardh	CB	–	–	–	–	–	–	–	+
C. serrulata (Forsskål) J. Agardh	CB	–	–	–	–	–	–	–	+
C. sertularioides (S.D. Gmelin) M. Howe	CB	–	–	–	–	–	–	–	+
Halimeda macrophysa Askenasy	C	–	–	–	–	–	–	+	–
Penicillus sibogae A. Gepp & E.S. Gepp	C	–	–	–	+	–	–	–	–
Pseudochlorodesmis furcellata (Zanardini) Børgesen	F	–	–	–	–	–	–	+	–
Rhipidosiphon javensis Montagne	C	+	++	+	+++	–	–	+	++
Bornetella sphaerica (Zanardini) Solms-Laubach	C	–	–	–	–	–	–	–	+
Neomeris annulata Dickie	C	–	–	–	–	–	–	–	+
Acetabularia dentata Solms-Laubach	C	–	–	–	–	–	–	–	+
Parvocaulis polyphysoides (P.L. Crouan & H.M. Crouan) S. Berger et al. [=*Acetabularia exigua* Solms-Laubach]	C	–	–	–	–	–	–	–	+
OCHROPHYTA									
Kuetzingiella elachistaeformis (Heydrich) M. Balakrishnan & Kinkar [= *Ectocarpus elachistaeformis* Heydrich]	FT	–	–	+	+	–	–	–	–

Table. (Contd.)

List of species	Morpho-functional forms	1st month, March		3rd month, May			6th month, August	55th month, March	60th month, May
		SE	AE	SE	AE	SpE	SpE	DbS	DbS
Ectocarpus sp.	FT	+	+	+	+	+	–	+	–
Feldmannia irregularis (Kützing) Hamel	FT	–	–	+	+	++	–	+	+
Hincksia mitchelliae (Harvey) P.C. Silva	FT	+	++	++	+++	–	+	+	+
Chilionema ocellata (Kützing) Kornmann	FT	–	–	–	++	–	++	–	+
Sphacelaria novaehollandiae Sonder	F	–	+	++	+++	–	+	++	++
S. rigidula Kützing	F	–	–	+	+++	–	–	–	+
*S. tribuloide*s Meneghini	F	–	–	–	++	–	–	–	–
Lobophora variegata (Lamouroux) Womersley	Pl	–	–	+	+	–	–	+	++
Dictyota friabilis Setchell	PlB	–	–	+	+	–	–	++	+
D. humifusa Hörnig, Schnetter & Coppejans	PlB	–	–	–	–	–	–	+	–
D. linearis (C. Agardh) Greville	PlB	–	–	–	–	–	–	+	–
Dictyota sp.	PlB	–	–	–	–	–	++	–	+
Chnoospora implexa J. Agardh	CB	–	–	+	+	–	–	–	–
Padina boryana Thivy	L	–	–	–	–	–	–	+	+
Turbinaria ornata (Turner) J. Agardh	CB	–	–	–	–	–	–	+	+
RHODOPHYTA									
Chroodactylon ornatum (C. Agardh) Basson	MF	–	–	–	–	–	–	+	–
Stylonema alsidii (Zanardini) K.M. Drew	MF	–	+	–	+	+	+	++	++
Erythrotrichia carnea (Dillwyn) J. Agardh	MF	+	+	++	++	+	++	++	–
Sahlingia subintegra (Rosenvinge) Kornmann [= *Erythrocladia subintegra* Rosenvinge]	M	–	–	–	–	–	–	+	–
Acrochaetium catenulatum M.A. Howe	MF	–	–	–	–	–	+	–	–
A. gracile Børgesen	MF	–	–	–	–	–	+	+	–
A. hypneae (Børgesen) Børgesen [= *A. seriatum* Børgesen]	MF	–	–	–	–	+	+	–	+
A. microscopicum (Nägeli ex Kützing) Nägeli [= *A. crassipes* (Børgesen) Børgesen]	MF	–	–	–	–	–	–	+	–
A. subseriatum Børgesen	MF	–	–	–	–	–	–	+	–
A. virgatulum (Harvey) Batters	MF	+	+	+	+	+	–	++	–
Acrochaetium sp.	MF	–	–	++	+	–	–	+	++
Hydrolithon farinosum (J.V. Lamouroux) D. Penrose & Y.M. Chamberlain	CC	–	–	++	++	++	+	++	++
Jania adhaerens J.V. Lamouroux	SC	–	–	++	+	–	–	+	–
J. capillacea Harvey	SC	–	+	++	++	++	+	–	++
J. ungulata f. brevior (Yendo) Yendo	SC	–	–	+	+	–	–	–	–
Amphiroa fragilissima (Linnaeus) J.V. Lamouroux	SC	–	–	–	–	–	–	–	+
Pneophyllum fragile Kützing	CC	–	–	++	+	+	–	+	–
Gelidium pusillum (Stackhouse) Le Jolis	T	–	–	–	–	–	–	+	–
Gelidiella acerosa (Forsskål) Feldmann & Hamel	CB	–	–	–	–	–	–	–	+
Parviphycus adnatus (E.Y. Dawson) B. Santelices [= *Gelidiella adnata* Dawson]	T	–	–	–	–	–	–	–	+
P. tenuissimus (Feldmann et Hamel) B. Santelices [= *Gelidiella pannosa* (J. Feldmann) J. Feldmann et G. Hamel]	T	–	–	–	–	–	–	+	+
Asparagopsis taxiformis (Delile) Trevisan de Saint Leon [= *Falkenbergia hillebrandii* (Bornet) Falkenberg]	F	+	+	+	+	+	–	+++	++

Table. (Contd.)

List of species	Morpho-functional forms	1st month, March		3rd month, May			6th month, August	55th month, March	60th month, May
		SE	AE	SE	AE	SpE	SpE	DbS	DbS
Hypnea sp.	CB	–	–	–	–	–	–	+	+
Peyssonnelia conchicola Piccone & Grunow	Cr	–	–	–	–	–	–	–	+
Peyssonnelia sp.	CC	–	–	–	–	+	–	+	–
Ceratodictyon spongiosum Zanardini		–	–	–	–	–	–	+	–
Champia parvula (C. Agardh) Harvey	CB	–	–	+	+	–	–	+	+
Ch. vieillardii Kützing	CB	–	–	–	–	–	–	+	–
Gelidiopsis intricata (C. Agardh) Vickers	CB	–	–	++	++	–	+	++	+
Lomentaria corallicola Børgesen	CB	–	–	–	–	–	–	–	+
Coelothrix irregularis (Harvey) Børgesen	CB	–	–	–	–	+	–	+	+++
Anotrichium tenue (C. Agardh) Nägeli	F	–	–	–	+	–	+	–	+
Centroceras clavulatum (C. Agardh) Montagne	MF	–	+	++	++	+	+	++	++
C. inerme Kützing	MF	–	–	+	+	–	–	–	–
Ceramium fastigiatum Harvey	MF	–	–	–	++	–	–	+++	–
C. fimbriatum Setchell & N.L. Gardner	MF	–	–	–	–	–	+	–	+
C. flaccidum (Harvey ex Kützing) Ardissone	MF	–	+	++	+	+	+	+	+
C. procumbens Setchell & N.L. Gardner	MF	–	–	–	–	–	–	–	+
Ceramium sp.	MF	–	–	+	+	–	–	++	+
Corallophila apiculata (Yamada) R.E. Norris [=*Centroceras apiculatum* Yamada]	MF	–	–	+	+	–	++	+	–
Crouania attenuata (C. Agardh) J. Agardh	F	–	–	–	+	+	–	–	+
Griffithsia metcalfii C.K. Tseng	F	–	–	+	+	–	–	+	–
Spermothamnion sp.	MF	–	–	–	–	–	+	+++	+
Spyridia filamentosa (Wulfen) Harvey	F	–	–	–	+	–	+	+	++
Heterosiphonia crispella (C. Agardh) M.J. Wynne	MF	–	+	–	+	+	–	–	–
Martensia fragilis Harvey [= *M. pavonia* (J. Agardh) J. Agardh]	MFl	–	–	–	–	–	–	+	–
Taenioma perpusillum (J. Agardh) J. Agardh	CB	–	–	+	–	–	–	–	–
Chondria dasyphylla (Woodward) C. Agardh	CB	–	–	–	+	–	–	–	–
Chondria sp.	CB	–	–	–	+	–	+	–	–
Digenea simplex (Wulfen) C. Agardh	CB	–	–	–	–	–	–	–	+
Herposiphonia secunda (C. Agardh) Ambronn [= *H. tenella f. secunda* (C. Agardh) Wynne]	MF	–	–	+	+	+	–	+	+
H. secunda f. tenella (C. Agardh) M.J. Wynne	MF	–	+	+	+	+	+	++	+
Laurencia carolinensis Saito	CB	–	+	++	+	+	–	–	+
L. intricata J.V. Lamouroux [=*L. implicata* J. Agardh]	CB	–	–	+	+	–	–	–	–
L. obtusa (Hudson) J.V. Lamouroux	CB	–	–	++	+	+	+	+	++
Laurencia sp.	CB	–	–	+	+	–	–	+	+
Polysiphonia scopulorum var. *villum* (J. Agardh) Hollenberg [= *Lophosiphonia villum* (J. Agardh) Setchell & Gardner]	F	–	–	+	+	+	+	+	–
Polysiphonia sp.	F	–	+	++	++	–	–	+	++
Tolypiocladia glomerulata (C. Agardh) F. Schmitz	F	–	–	–	–	++	–	–	–

SE and AE—experiments in the sea and in aquaria, respectively, corals were damaged by a handsaw; SpE—an experiment with coral pebbles in the sea; DbS—coral damage after bleaching in 1998. Morphofunctional algae forms: CB—coarsely branched, S—sponge-like, C—calcareous, Cr—crusted, CC—crusted coralline, L—leaf-like, P—pulpy, M—microscopic, B—bag-like, MF—microfilamentous, F—filamentous, FT—filamentous tubular, Pl—plate-like, PlB—plate-like branched, MFl—meshed film-like, Pt—turf-like, FF—fine filamentous, SC—segmented coralline, En—endolith. Occurrence of dominating morphofunctional forms: +++ more than 100 spec. per 1 cm^2 of such forms as M, MF and FF; ++ more than 100 spec. per 1 cm^2 of F, FT, P and Pt forms; + more than 1 spec. per 1 cm^2 of C, S, CB, CC, L, B, Pl, PlB and SC.

experiments diatomic and green-blue algae that settled on the dead wounded parts of mechanically damaged colonies made up more than 50% of all settlers, whereas on coral pebbles they constituted only 20%. The projective coverage of wounds with algal settlers was always higher than that of coral pebbles, cement and concrete blocks [146]. The substrate of the dead part of a wound significantly differs from artificial substrate and dead coral colonies by the presence of a thick, loose mucus layer [42] and other excretions of the coral's living tissue. Most likely these excretions serve as a convenient substrate for algal colonization, and some of them (dissolved in water) can be assimilated by algae in the process of organotrophy [3, 4]. In our experiments the first algal settlers on mechanically damaged corals included *Nitzschia longissima* (Bacillariophyta), *Leptolyngbya tenuis (= Phormidium tenue), Calothrix* sp. (Cyaniobacteria), *Ulva* (= *Enteromorpha*) *clathrata* (Chlorophyta), *Hincksia mitchelliae* (Ochrophyta), and *Ceramium flaccidum* (Rhodophyta). Wounds from other damage (for example, by a thin water sheet) were inhabited during the first experimental month with the same species of diatoms, blue-green, and green algae, as well as with *Ulva* (= *Enteromorpha*) *compressa* (Chlorophyta) and *Spyridia filamentosa* (Rhodophyta) [see 147]. After the mass loss of corals in 1998 dead coral colonies on the Great Barrier Reef of Australia were colonized by pinnate diatoms, blue-green (*Plectonema terebrans* and *Spirulina subsalsa*), green (*Cladophora dalmatica* and *Enteromorpha linza*), brown (*H. mitchelliae* and *Sphacelaria* spp.), and red (*Gelidiella pannosa* and *C. clavulatum*) algae [43].

The first algal settlers occupying the substrate change its relief and promote the accumulation of organic and inorganic sediments on the wound surface [45, 114] (figure, I); algal colonies and thalli facilitate propagula attachment and further obliteration of wounds with epilithic macroalgae and epiphytes settled on them. Thus, algal first settlers, fastened to a newly formed substrate in the first month of colonization, do not impede the settling of the other algal species, and, on the contrary, facilitate it.

In experiments organized in the sea, during the first and third months of colonization, the number of algal species increased, reaching 55 species on wounds and 34 species on coral pebbles (see the Table). At the same time, some algal first settlers disappeared. Within the next three months of colonization the succession of algal communities continued, but on wounds the total number of algal species was reduced due to disappearance of the algal first settlers, whereas on coral pebbles it rose to 48 species (see the Table). Among newly appearing algae lamellar *Lobophora variegata* (Ochrophyta) and filiform *C. clavulatum* algae, as well as calcined incrustations of the epiphytic alga *Pneophyllum fragile* (Rhodophyta) dominated on coral pebbles (see the Table). Monodominant communities of *L. variegata*, calcined corticated *Peyssonnelia conchicola, Litophyllum* sp. and branched *Hypnea spinella* (Rhodophyta) were also recorded.

During a six months succession a community of algal turf was formed on wounds of mechanically damaged corals and on coral pebbles. This was established by comparing the species composition and dominant algal species on wounds and pebbles in the experiment and on coral colonies damaged in July–August of 1998 (55 and 60 months after the damage) (see the Table). In the both cases turf-forming algal species, such as *H. spinella, Jania* spp., *Laurencia* spp., *C. clavulatum, Ceramium* spp., *S. filamentosa, Herposiphonia secunda, Polysiphonia* spp., *Coelothrix irregularis* (Rhodophyta), *Dictyota* spp., *Sphacelaria* spp., (Ochrophyta), *U. clathrata, U. compressa,* and *Cladophoropsis* sp. (Chlorophyta) dominated.

Algal community formation on coral pebbles slightly differed from that of damaged corals, although the rate of succession was lower. For example, the algal projective cover, the total number of algal species, and the composition of dominating species of the 6-month community on coral pebbles were close to those of 3-month algal communities on the wounds of mechanically damaged colonies. The difference in succession rate can be explained only by differences in the physical and chemical properties of those substrates.

Thus, algal colonization of wounds of mechanically damaged corals and coral pebbles corresponded to the typical process of formation of algal turf with the predominance of diatomic, blue-green and green filiform algae on the early stages of succession, and green, brown and red algae (with more complicated morphologies and greater size) on the following developmental stages. Such a tendency for algal community formation was observed earlier on dead and damaged coral colonies of the Great Barrier Reef of Australia [43], on coral reefs of the Atlantic [83–85] and the Pacific Ocean [120].

Algal diversity and composition in newly formed turf communities depended on the season. For example, the number of species in algal turf on old wounds was 25% higher in the summer than in the winter (see the Table). In the summer period the species composition of algal turf became more diverse, mainly due to the appearance of short-lived algae (often epiphytes): diatomic (*Nitzschia* spp., *Navicula* sp., *Tabellaria* sp.), blue-green (*Lyngbya majuscula, Phormidium* sp., *Spirulina major, S. subtilissima, S. tenerrima, Calothrix contarenii, Calothrix* sp., *Dichothrix* sp.), brown (*Sphacelaria rigidula, Chilionema ocellata, Ectocarpus* sp.), and red (*Taenioma perpusillum, Ceramium procumbens, C. fimbriatum*).

Experiments on algal community formation on various types of wounds have been conducted in the sea and in aquaria under similar environmental conditions (light, temperature, nutrients), but in the absence of herbivorous animals in aquaria they showed that algal composition and abundance in the formed communities

depended on the conditions of coral maintenance and only to a slight degree on the type of wound [146, 147] (see the Table). In aquaria wounds algae appeared earlier on artificially inflicted wounds and formed denser populations with a greater (approximately 1.5 times) species number than in the sea. Moreover, some species that are rare in the sea, for example, *N. longissima, Derbesia marina, Rhipidosiphon javensis* (Chlorophyta), *S. rigidula,* and *C. flaccidum*, dominated in aquaria. *Fragillaria* sp. (Bacillariophyta), *Aphanocapsa littoralis* (Cyanobacteria) and other species were found only in aquaria. Such an algal composition in aquaria and in the sea allows us to suggest that the absence of herbivorous animals in artificial conditions provides an opportunity for many algal species to reach the reproductive stage and to germinate. Many filiform algae in the sea cannot develop reproductive organs due to permanent fish attack [18-20].

The effects of herbivorous animals on coral reefs was more than the ones discussed in publications devoted to restoration and degradation of coral reefs [13, 22, 131, 133]. According to the general opinion of researchers, herbivorous fish and invertebrates restrain colonization (or populating) of substrate with algae and overgrowing of coral polyps with them.

The influence of light intensity on algal community formation, both on damaged coral wounds and on coral pebbles affected the number of algal species that settled on these substrates at various succession stages. At early succession stages (the first month), when keeping samples under the conditions of bright (90–70% of photosynthetically active radiation (*PAR*), falling on the water surface) and moderate (30–20% *PAR*) light no major differences in species number were found. At the later stages (in 3 months and later) both on wounds and on pebbles the number of algal species under the moderate light was much greater than under strong and weak (5–2% *PAR*) lights. The number of algal species on old wounds (1998) in a developed turf community did not depend on light intensity within the range of 20–90% *PAR*, but under weak light species diversity was much less. It is known that the absolute majority of algal species has a wide light range of acclimation (from 90–80 to 2–1% *PAR*). The ability of algae to adapt to a wide light range determines the high species diversity and population density of algae in a turf community [5].

It was shown earlier that some algae can grow in the intertidal zone under 100% *PAR*, as they are more resistant to bright light due to accumulation of pigments–protectors in their cells [10, 160]. At the same time, some red algal species are known (mainly calcined incrustation ones) which are able to adapt to a less than 1% *PAR* [7]. We propose that precisely these algal properties determine a shift in the direction of succession of their communities that were developed under the conditions of high and low light. It is necessary to note that the green algae *Monostroma nitidum, Ulva pertusa*, and *U. compressa* are dominant under high light, whereas such red algae as calcines crustacean *P. conchicola*, roughly branched *Gelidiopsis intricata*, filiform *C. clavulatum* and *Polysiphonia* sp. dominate under low light.

Thus, already in the first year after a catastrophe abundant obliteration of a newly formed substrate with benthic algae takes place (extensive phase shift), at which algae cover 60–90% of the bottom and corals only 5–30% [43–45]. However, not all reefs are colonized by macrophytes after a catastrophe. Microalgae, making "mats," red calcareous crustacean algae, sponges and other animals settle on a free substrate, forming dense populations [25].

"A phase shift" on a coral reef, or reef obliteration with macrophytes, can be a temporary phenomenon. Coral populations can be restored to their original or modified shape, and "a vegetable reef" can become a coral reef again. But such an optimistic prognosis cannot be true for every damaged reef, as coral population restoration depends on many factors, where the key ones are as follows: 1) the degree of damage of a coral reef; 2) the presence of a source of coral planulae near the damaged reef (i.e., living colonies of corals lost on this reef); 3) the composition and abundance of algal settlers competing with corals for a newly formed substrate, space and resources; 4) outer environments facilitating or impeding restoration of the coral reef.

A pessimistic prognosis for a damaged coral reef can include its further degradation or its transformation into a vegetable reef, as well as the cessation of calcium carbonate sedimentation in a reef basis and destruction of this basis by plants, animals, bacteria and storms. All this leads to the replacement of a diverse and highly productive coral reef ecosystem by a very poor ecosystem of loose sandy earth, or, at best, of sea-grass bush.

It is necessary to note separately that transformation of a coral reef damaged as the result of natural phenomenon into a vegetable one and its further destruction is usually accompanied by the negative impacts of human activity. As Jokiel et al. remarked [73], coral reefs are able to be restored after natural catastrophes, but not after catastrophic water pollution by anthropogenic effects. Reefs damaged by natural cataclysm, exposed to the effects of environments polluted with organic and inorganic substances from the fluid wastes of large cities and villages, various dredges, and agricultural effluents rich in fertilizers, pesticides and other inhibitors lose their ability to recovery. Weakened coral colonies become overgrowth with aggressive algal species, such as, for example, *Dictyosphaeria cavernosa* (Chlorophyta), *Hypnea pannosa* (Rhodophyta), *L. variegata*, with cyanobacteria and others, which transforms a coral reef into a vegetative one, and dooms it to destruction [47].

One of the main conditions for coral reef restoration is the victory of polyp colonies of different coral species in competition with algal settlers, sponges and soft

corals for substrate, space, and resources. This, in its turn, depends on the mechanisms and rates of restoration of coral polyps on a substrate free or occupied by new settlers.

Restoration of Coral Polyps on Damaged Colonies

Depending on the damage type, wounds on coral colonies can be deep or surficial, and can contain or not contain living tissue. Thus, Hall [56, 57] distinguished four main types of wounds on damaged corals: 1) partial loss of soft tissue, eaten by sea-stars and gastropods; 2) the loss of soft tissue with its subsequent separation from a skeleton upon coral bleaching under the effects of high or low water temperature; 3) the partial loss of soft tissue and skeleton as the result of their removal by fish or solid objects; 4) a substantial loss of a part of the skeleton together with the soft tissue under the effects of storms and cyclones, as well as explosions and other human activity. Wounds of all types, caused to corals by natural or human activities, can be healed with living tissue, and colonies are able to be restored. The rate of wound healing depends on their nature, size, the physiological state of the damaged colony, as well as on external abiotic and biotic factors [56, 92, 96-98, 100–102, 118, 146–148].

The mechanisms for restoration of coral tissue and skeletons have been described in a number of papers [14, 56, 57, 92, 98, 101, 118, 147, 148]. It was shown that there are three stages of wound healing. At the first stage damaged soft coral tissue is restored. This is the quickest process, which lasts several days after the damage and depends only on the physiological state of the coral. At the second stage restored soft coral tissue "creeps" on the substrate (the dead part of a wound) and construct new polyps on it (figure, I and J). The rate of coral regeneration at this stage depends first of all on the substrate's state (its physical and chemical properties), on the physiological state of the coral, as well as on the external abiotic (light, temperature, and sedimentation) and biotic (composition and abundance of algae and animals-settlers on the wound surface and the presence and abundance of phytophagan fish) factors. This regeneration stage lasts up to 6–8 months and ends with the end of the creeping of coral tissue on the substrate due to formation of an insurmountable obstacle in a form of a thick community of algal turf (figure, G, H, L). The rate of occupation of the dead wounded part by new polyps at this stage is from 2 to 10 mm per year for different coral species [146]. The third stage of regeneration of damaged colonies starts after the formation of an algal turf with large macrophytes and benthic animals on a wound, which stops the advance of coral tissue along the wound surface. This regeneration stage lasts up to the complete healing of a wound with coral polyps, or up to the minimization and localization of a part of a colony's dead skeleton, occupied by algal species or animals, which are aggressive to the coral (figure, L).

At every stage regeneration is performed by special mechanisms. At the first stage damaged tissue is quickly restored (3–5 days), and damaged polyps regenerate over 10–20 days, at that time a clear border is established between the living and dead (coral skeleton) wounds (figure, I). Comparing the rate of coral tissue regeneration at this stage of six coral scleractinian species, we found out that species-specific differences of this parameter depended only on the proportion of damaged and dead wound surface, which, in its turn, depended on morphological and anatomical properties of a coral [146].

Mechanisms of polyp regeneration on the dead part of a coral at the second stage (figure, J, K, M, N, O) substantially differ. We have established that various morpho-functional forms of *Porites* genus corals have the same restoration mechanisms, whereas *Acropora* and *Pavona* genera species have different ones. Coral tissue of *Porites* genus species, which practically don't containing calcium carbonate crystal (skeleton), "crept" on the dead part of a wound as a 50–200 μm band, adjoining it closely, and covering or "flowing" around substrate burrs, algae and animal settlers (figure, J and M). When touching living and dead surfaces coral excretes calcium carbonate crystals and entombs living organisms in its skeleton (figure, K). The maximal rate of tissue advance along the dead wound surface, measured for four coral species, was 0.24 ± 0.11 mm/24 hours. This restoration mechanism was typical for massive and branching coral colonies.

In the *Acropora* and *Pavona* genera the dead wound was covered by a thin hard "lamina" consisting of calcium carbonate crystals and covered by soft tissue. Algae and animals, covered by such lamina, were killed, being entombed in the coral skeleton (figure,K). The maximal rate of lamina advance over the substrate was 0.54 ± 0.17 mm/24 hours. This mechanism is used mainly by incrusting and leaf-like corals. The rate of coral regeneration at the second stage weakly depended on light intensity and water temperature as distinct from the growth rate of undamaged coral colonies. Most probably restoration of the damaged colony sites requires substances from its healthy parts together with photosynthetic products of zooxanthellae from the neighboring polyps, in this case not only products of zooxanthellae photosynthesis are used, but also products of predation and other sources of coral nutrition.

At the third stage of a colony restoration (experimental investigations were not carried out), according to our observations, after ceasing coral tissue advancement along the dead wound surface, polyp gemmation and growth on the damaged and restored tissue did not stop; new polyps began to develop at a 2–3 mm distance from the dead wound. A peculiar spindle of young growing corallites (polyps) was formed. The spindle was gradually converted into a "peak" over the old wound. In one–two years or in several years, oppositely directed peaks grew together (figure, L). Plants

and animals–settlers covered by the peak were killed, and their remains were entombed in the coral skeleton. As it was shown by Glynn and Fong [53], coral colonies grow intensively during the regeneration process, and their total surface and tissue weight increase. Thus, during a year the size of damaged colonies of *Pocillopora elegans* coral increased approximately by 40%, whereas that of *Pavona clavus* grew by 20%. The high growth rate of the damaged corals could be connected with the process of polyp regeneration, which stimulates growth at the third stage of wound obliteration.

Competitive Interrelation of Coral Polyps And Algae-settlers on a Damaged Coral Reef

McCook et al. [95] list 13 papers in their review, in which the competitive interrelation of algae and corals have been investigated with indication of the victor [15, 40, 60, 65, 74, 94, 96, 99, 101, 103, 104, 137, 153]. We can add several more papers to this list: [75–77, 145–148]. The negative effects of algal communities on coral vital functions are described in 12 of these 20 papers, the positive effects of epilithic algal communities on coral growth and production are described in three works, whereas seven works describe the negative effects of corals on neighboring algal communities. This simple analysis of the published data shows that on damaged coral reefs in the state of a "phytogenic shift" [47, 66, 80, 83, 84, 103] the victory of algae in competition for space and resources is not unequivocal, and the fate of a coral reef is determined by the competitive abilities of both algae and corals in particular habitat conditions. In their turn, the competitive abilities of organisms depend on the presence of mechanisms of competitive inhibition and physiological state of a competitor in particular environmental conditions.

Competition for light between corals and algae is most likely possible only at a large depth or with strong shading, as the range of light tolerance of algae and corals is wide enough and close for both groups of organisms [7]. It is necessary to note that hermatypic corals are more adjusted to dwelling at great depths and with strong shading than algae, as they have five modes of feeding, four of which are heterotrophic ones [8, 9], whereas algae mainly have only one, the autotrophic mode of feeding.

Competition between algae and corals for carbon, nitrogen and phosphorus is improbable due to the fact that algae and corals have different sources of these biogenic elements [8, 9, 108, 109, 123, 124]. Thus, coral zooxanthellae photosynthesize at the expense of metabolic carbon dioxide and bicarbonate (CO_2, liberated by animal, viz., coral polyp, respiration), whereas epilithic animals do so via CO_2 and bicarbonate dissolved in water. Corals obtain organic nitrogen and phosphorus mainly by means of predation or filtration of bacteria and organic remains, as well as from dissolved organic substances, whereas mineral substances dissolved in water are sources of nitrogen and phosphorus for epilithic algae.

McCook et al. distinguished the following common mechanisms of competitive inhibition for algae and corals: 1) overgrowth, or direct interaction and physical deprivation of all vital resources; 2) shading, an indirect effect, limiting or depriving a competitor of opportunity to use light, one of the main sources of energy; 3) abrasion, a direct physical effect, expressed in the damage of the external tissues of a competitor; 4) chemical effect, or allelopathy, a direct chemical effect, expressed in evolution of toxins, inhibiting the photosynthesis, growth, or reproduction of a competitor; 5) occupation of a substrate and space, creation of a recruitment barrier; 6) protection against the competitor's impact. Both corals and algae possess such competitive mechanisms. But possession of one or another mechanism is not the same as victory in the competition, i.e., weakening a competitor or forcing it out of a community. The efficient use of a mechanism depends on the physiological state of a competitor, which in its turn is affected by external conditions, i.e., various abiotic and biotic factors. Thus, for example, frequent storms impede the settling of coral larvae (turning over, abrasion, and carry-over of moving substrate, viz., coral colony debris or coral pebbles) [107]. Dredging and settling on a solid substrate also prevent the attachment of algal spores and coral larvae [16, 17, 126]. Animals, eating young coral colonies [48, 128–130] and fish, eating algae, do not promote victory in the competition for substrate [62].

Overgrowth of competitor or its use as a substrate. This most prevalent competition mechanism is used both by algae and coral. If competitors have no chemical or physical protective devises, then one of them, when engaging the other one, overgrows it and deprives it of all resources, and uses it as a substrate (figure, J, L, M, P). Based on the example of the first stages of wound obliteration in damaged corals of Okinawa we showed that coral polyps are able to overgrow more than 100 species of diatoms, blue-green, green, brown and red algae, and only some blue-green algae (*Lyngbya majuscula, L. semiplena*) present a chemical obstacle for them (figure, N and O). For about 3–4 months algae that settled on the surface of coral wounds did not present a serious obstacle for polyps due to the small size of the algae (figure, I and J). Coral polyp advance along a newly formed substrate continued until a thick algal turf layer that formed on the wound surface became a physical obstacle for coral tissue advance. Upon the long-term or permanent presence of such an algal community polyps situated on the border with the algae under unfavorable conditions for coral can be lost and overgrown by algae. However, in conditions comfortable for coral, colony wounds more often are overgrown by coral polyps, i.e., corals conquer the algae in the competition for substrate.

It has been shown more than once that algal turf does not present a serious obstacle for the growth and restoration of partially damaged coral colonies already obliterated by algae [15, 96, 99, 101, 102, 146, 154], able to overgrow the turf (see above). At times algal turf contains blooming blue-green algae, which are toxic for corals and present an insurmountable chemical obstacle for them. At the same time, algae toxic for corals are rather rare, and they are mainly ephemerae. Having survived their presence on the border with substrate, coral rehabilitates itself and continues its attack against algal turf [146]. Herbivorous fish and invertebrates, which eat away young thalli or certain algal species, for example, Pomacentridae, and *L. majuscula*, which is toxic to coral [see: 32, 55, 82] facilitate the overgrowth of algae by corals. However, corals don't always conquer algal turf. Under the extremely low light intensity, especially in the lower colony parts, coral polyps become weak and lose the ability to entomb algal spores and fragments in the skeleton. In such conditions the red algae *C. clavulatum* and *Polysiphonia* sp., as well as crustacean algae settle on still-living coral and form a thick bush.

Shading. Shading as a mechanism of competition (as well as overgrowth) is directly connected with the growing activity of a competitor. One organism partially or fully deprives its competitor of one of its main energy resources—light, while not being in direct contact with it. This competitive mechanism is used both by corals and for algae. As was shown above, incrusting coral species growing over a substrate as a peak shade the algae settled on the substrate and then entomb them in their skeleton (figure, K). At the third stage of the restoration of damaged colonies the algae that are settled on the wounds are completely deprived of light (figure, L). It can be hypothesized that it is precisely the shading of the bottom by coral that facilitates the settlement of red crustacean algae, which do not require high illumination, under the coral, but not of thalline macrophytes, which are able to compete with corals for substrate and space. However, this competition mechanism is usually used just by large macrophytes such as *Sargassum* spp. or *Hypnea* spp. There are some data testifying that in shallow waters shading of corals by macrophytes protects them from bleaching and loss. It has been shown that the planulae of corals from *Porites, Favia* and *Favites* genera survive better in algal turf than on bare substrate [24]. According to the data of Suzuki and Hayashibara [136], the planulae of corals from the *Acropora* genus do not attach to a substrate and do not survive at an illumination more than 500 μmol/(m^2 c), at an illumination of 150 μmol/(m^2 c) more than 50% of the larvae attach and survive, and in full darkness about 65% survive.

The effects of shading on the morphology and physiological state of corals have been well-studied [9]. It has been shown that the majority of scleractinian coral species are able to adapt to an illumination from 100 to 1% *PAR* [7]. Within the rage of illumination from 90 to 3% *PAR* light adaptation of coral polyps occurs without the loss of production resources [144, 145], whereas the use of these resources within the light range of 80–20% *PAR* provides permanently high and stable production level [139]. It has been shown that such corals as *Porites lichen, P. australiensis, Favites abdita, Alveopora spongiosa, Favia stelligera, Pavona decussata,* and *Leptastrea bewickensis* dwell in the niches and grottoes of the coral reefs of the South China Sea islands at an illumination of 0.3–0.5% PAR_0, which is approximately from 4 to 7 μE/(m^2 c) in the middle of a sunny day [141]. For some coral species approximate lower limits of their light adaptation were found: for *S. pistillata, Porites attenuata* and *Echinopora lamellosa*, viz., a 0.1% *PAR*, for *Pocillopora damicornis*, a 0.8% *PAR* for *Seriatopora calendrum*, and a 2% *PAR* for *S. hustrix*. In a long-term (6 months) experiment coral adaptation to extremely low light (from 0.1 to 2% *PAR*) was continued from 80 to 120 days. In this case the population density of coral zooxanthellae was reduced considerably (to 5–20% as compared to the initial density), and, accordingly, primary production as well [145]. At the same time (as was shown for *S. pistillata*), under the condition of extreme shading, coral polyps begin killing and digesting zooplankton [143]. WIth shading *Galaxea fascicularis* optimizes its heterotrophic feeding through modification in the colony's morphology [41]. Based on the obtained data we think that algal shading of corals can negatively affect their physiological state only in extremely shaded locations. On coral reefs such shading can be provided by Sargassum algae bush. Thus, with 100% projective cover of the bottom by Sargassum algae in the upper sublittoral 2–2.5 m depth, (Thu Island in the South China Sea) and bottom illumination of 1–2% *PAR* coral was shown to be absent in Sargassum algae bush [140].

Abrasion or a direct physical impact. Coral polyps can be damaged by algae, having rough thalli (*Sargassum* spp.) or pointed hard tenons (*Turbinaria* spp.), especially during storms. However, algal abrasion of corals has not been shown experimentally [95], or recorded in nature. It is possible that such damage is quickly healed, and the coral is restored during intervals between storms. In the case of frequent or permanent contact of a coral surface with algae of any form, which can be observed on the border between polyp and algae communities, the latter impede the opening and movement of polyp tentacles when catching zooplankton, and in that way impair their physiological state. If this continues for several months, polyps become weak and perish, whereas algae occupy the free substrate. Thus, the competitive mechanism of temporary direct physical contact (but not of abrasion) precedes the effect of the other mechanism of competitive relations—coral overgrowth with algae. Such a mechanism for negative effects on corals is used for the most widely distributed algal community of coral reefs—algal turf.

Allelopathy, or the mechanism of chemical impact. Sea algae and corals excrete various organic substances into water, many of which are biologically active [3, 4, 6]. It thought that some algal species growing near coral colonies or directly on them, negatively affect the physiological state of these corals, hypothetically in an allelopathic manner, i.e., through the excretion of biologically active substances into the water [95, 111, 116]. The chemical impact of algae on coral through the water column has not been proved thus far experimentally [95], but it has been established when in direct contact. Thus, it has been revealed that the fine-branched red alga *Dasyopsis spinuligera*, obliterating coral colonies in the Caribbean Sea kills corals by excretion of toxic substances [86]. Red algae of the Ceramiaceae family (*Corallophyla huysmansii* and *Anotrichium tenue*) can attach themselves to living corals and gradually kill their polyps by excreting toxic substances into the water [76, 77]. Both in the first and in the second case the effects of algal toxic substances on corals are only suspected. The study of de Nys et al. [115], indicating that the red alga *Plocamium hamatum*, being in direct contact with the octocoral *Sinularia cruciata*, causes coral tissue necrosis with the effect of the second algal metabolite chloromertensene, is a demonstration of a negative allelopathic effect of algae on corals.

We more than once have observed bleaching of *Porites* genus coral polyps from the border with a community of the blue-green alga *L. majuscula* [146](figure, N). Another species—*Lyngbya semiplena*—stopped tissue advance along a dead wound and regeneration of polyps of *Porites cylindrica* coral (figure, O). The allelopathic effect of *Lyngbya bouillonii* on *Porites lutea* coral was studied in an experiment. During 30 days of direct contact Lyngbya caused coral bleaching up to almost a complete loss of zooxanthellae. At the same time, in this experiment *Dictyota dichotoma* did not cause negative changes in the physiological state of the coral, and *P. lutea*, on the contrary, suppressed the growth of *D. dichotoma* thalli [150]. Although we did not determine what toxic substances were excreted by *L. bouillonii*, nevertheless the allelopathic effect of this blue-green alga on the coral was evident. It is known that some Lyngbya species, for example, *L. majuscula*, produce such biologically active substances as curacin, barbamide, dichlorbarbamide and others, which are poisonous for many animals and humans [2, 33, 51, 91, 159]. Several other cyanobacteria—*Nodularia, Oscillatoria* and *Gloeotrichia*—are also poisonous [2]. But species poisonous to coral species are rather rare among fouling algae. It has been shown that scleractinian corals are able to overgrow more than 100 various algal species from Bacillariophyta, Cyanobacteria, Chlorophyta, Ochrophyta, Rhodophyta departments, and only three Lyngbya species (Cyanobacteria) can stop the regeneration of damaged corals [146, 147, 150].

An allelopathic influence of competitors on each other takes place upon their direct contact. Thus, negative effects were not found from a winter and spring–summer algae community on 10 species of scleractinian coral when they were at a 1–2 cm distance from algal communities, [149]. At the same time, a spring–summer algae community increased the growth rate of *Pavona divaricata, Acropora hyacinthus* and *Acropora* sp. 2–4 times. Stimulation of a coral growth rate by algae can hardly be explained by an allelopathic effect. More likely these coral species can eat small animals inhabiting the algal community, which stimulates their growth.

Creation of a recruitment barrier to a competitior. Reservation, seizure, and unimpeded occupation of substrate and space is one of the main mechanisms of algae competition [5, 117], used on a damaged coral reef. Hermatypic corals do not have such a competitive mechanism due to their slow growth. As was mentioned above, substrate free for 6–8 months after a mass coral loss was inhabited by a dense diverse community of algal turf, which impedes the attachment of coral planulae to the substrate, as well as the development and growth of young colonies. A high thalli density and biodiversity, as well as the seasonal variability of species composition, presence of poisonous algal species in it, permanent refreshment of thalli, and consumption of young thalli by fish make this community practically unfit for coral to settle [16, 17, 38]. In this connection on damaged coral reef overgrown by algal turf, corals settle on areas free from algal turf: at the basis of coral colonies or large algae, on rhizoids of these algae, or on incrustations of red calcined algae. At the same time, Bena and Irikawa [24] often found planulae of corals from the *Porites, Favites* and *Favia* genera among algae overgrowing the substrate, and observed a high survival rate of these planulae in algal communities.

An algal turf community is assumed to be a permanent member of the coral reef community [121], but fairly often at some reef sites turf can be torn off by waves, large animals, humans or eaten by fish, i.e., free substrate permanently appears on a reef, and this can be occupied by young coral colonies. In this connection it is impossible to consider algal turf as the main competitor for substrate on a recovering coral reef. Moreover, algal turf does not present a serious obstacle for the growth of adult coral colonies, and can also serve as a source of living food (small animals dwelling in the turf) for corals and in this way stimulate their growth.

The area of bottom projective cover with large thalline algae is significantly larger (by dozens of times) than the substrate area occupied by rhizoids and other attachment organs. In this connection they cannot use the competitive mechanism of creation of a barrier for settlement of a competitor for substrate. As to red calcareous crustacean algae, they facilitate rather than impede the occupation of substrate by corals. It is known that coral planulae are easily attached to a cor-

alline algae surface, forming young colonies. However, the further fate of young colonies is not simple, as under a low light intensity (usually less than 5% *PAR*) coralline algae can overgrow coral polyps.

Protection Mechanisms

McCook et al. [95] proposed two examples of protection mechanisms against a competitor: 1) desquamation of the epithelium of red crustacean algae together with algal spores and animal larvae settled on their surface [70, 87], and 2) permanent clearing of an undamaged colony's surface of settled algal spores and fragments by means of renewal of the surface layer, consisting of mucus components.

Corals can protect themselves from algal spores and fragments using several other mechanisms, for example, by entombing them in their skeleton [146] (figure, P). Recently Nugues et al. [113] described an interesting protective mechanism of corals growing near algae. Corals suppress the growth of algae, touching them with their mesenterial filaments, in this case fragments of *Halimeda opuntia* green alga injured by nematocysts become bleached, and chloroplasts migrate from the injured segment to the other healthy ones.

We think that toxicity of algae is a mechanism of algal protection against corals. At present several algae species toxic to corals are known: primarily, they are blue-green algae of the *Lyngbya* genus [146–148, 150], as well as the red algae *Corallophila huysmansii, A. tenue* [see: 76, 77], *P. hamatum* [see: 115], *Dasyopsis spinuligera* [see: 86], and most likely the brown alga *L. variegata* [see: 75].

Importance of Competition in the Fate of a Damaged Coral Reef

Descriptions of coral reefs, the catastrophes of 3–10 years ago [12, 22, 25, 68, 69, 78, 132], and experimental works on the investigation of competition between algae and corals on variously damaged coral reefs [146–150] give us the opportunity to picture their further fate, which basically depends on the degree of coral damage, external conditions for coral survival after the damage, and the presence of a source of reproductive material for restoration of coral populations, as well as the biodiversity and abundance of new settlers—algae and animals. Renewal of a damaged coral reef starts immediately after a catastrophe starting from the restoration of injured colonies and the settling of a free substrate by young polyp colonies from planulae brought by currents from healthy reefs or living coral colonies of the damaged reef, and lasts until the complete recovery of its initial or modified shape.

We shall consider in detail the importance of one of the main conditions of restoration (or degradation) of a coral reef, namely competition between algae and corals on the damaged coral reef. Surviving coral colonies and young corals, appearing on the damaged reef after a catastrophe, start competition with new settlers (plants and animals). Competition for substrate starts at the first stages of colonization of a free substrate, newly formed after a catastrophe. The duration of this competition period is about one year. The main competition mechanisms for corals are overgrowing a competitor, and for algae, the construction of a barrier to competitor colonization and allelopathic effects. In this period on the damaged reef algae, the winners in the competition for substrate occupy its largest part. However, corals turn out more often to be the winners in the competition for substrate (wound surfaces) on the damaged reef.

After the formation of permanent algal communities on previously free substrate the competition between corals and algae continues, and a set of competitive mechanisms increases. The competition stage of quick and effective substrate seizure by algae turns into the stage of symbiotic relationships between individuals, if symbiosis is understood as the "joint living of unlike organisms" [1], exchanging signals [5]. Exchanging signals means any physical or chemical interaction. Symbionts (corals and algae) are connected with each other by neutral (i.e., living, not disturbing, and not helping each other), mutualistic (helping each other) and competitive interrelations. The symbiotic relationships of two or several species become competitive ones when some vital resource, space, substrate, light or mineral nutrition, become deficient. On a coral reef densely occupied by permanent algal associations and coral colonies photosynthetic (autotrophic) organism species compete mainly for substrate, space and light. At this stage of coral reef restoration corals display such competition mechanisms (which unfortunately have not been properly studied) as the allelopathic effect [150], direct physical effects, shading and protection against competitors, whereas algae use overgrowth, shading, direct physical effect and protection mechanisms. Not all algal species inhabiting a coral reef compete with corals. Epiphytes and ephemeras, which are dominant on a coral reef in their numbers of species, do not participate in competition. The main competitors for substrate are big thalline algae, algae toxic to coral, and dense permanent algal communities bordering coral polyps.

After formation of permanent algal communities on the damaged coral reef its restoration mainly depends on the competitive capacity of settlers (corals, algae, sponges, and others), environmental conditions on the reef and on the frequency and power of subsequent natural catastrophes. Recent investigations have shown that if a coral reef is populated after a catastrophe by large thalline macrophytes or certain sponge species, it will significantly hamper restoration of the coral reef [11], whereas its occupation by red coralline algae facilitates the attachment of coral planulae on their surface and restoration of the coral reef [25]. Under conditions of environmental pollution and other negative anthropogenic effects the coral reef is not restored [73].

Frequent occurrence of natural catastrophes delays complete restoration for an indefinite period of time.

Symbiotic relationships between settlers have a great importance for restoration of the coral population of a reef, as in the recently revealed symbiosis of the coral *Montipora digitata* and the sea grass *Thalassia hemprichii* [see: 112], when the sea grass, fixed by rhizomes to sandy bottom and parts of dead coral colonies, promotes the stability of this association to wave action, and also positively affects the growth of the coral colonies. In its turn, the coral somehow impedes settling epiphytes on the leaves of the sea grass. The coexistence of hermatypic corals and algal turf on neighboring areas is another example of symbiotic relations. As has been noted [24], algal turf does not hamper the settling and attachment of coral planulae and at the same time it protects them from predators. As well, polyps near an algal turf community hunt for small animals and larvae dwelling on the turf [149]. Red coralline algae occupy free substrate under conditions of moderate and weak light [146], and in their turn form a substrate for settling and attaching coral planulae [132, 136]. Large thalline algae shade corals (without direct contact) and protect them from bright light, increasing their production capacity [9]. Sargassum algae, occupying free substrate after coral loss on the outer side of a reef-flat and on a reef-flat, form an obstacle for storm waves, in this manner protecting the reef basis and young coral colonies developed on it from destruction, which facilitates its restoration.

A mutualistic relationship is typical for animal–polyp cells and endolithic green and blue-green algae. Using healthy colonies of *P. lutea* and *P. cylindrica* corals as a subject, and applying the methods of stable carbon and nitrogen isotope analysis, we have shown for the first time that vegetable and animal cells exchange metabolites containing nitrogen [148]. It has been also determined that endolithic algae and coral endosymbionts zooxanthellae use the same source of carbon for photosynthesis, viz., carbon dioxide excreted by polyp cells during respiration. It has been shown that when coral tissue is bleached, the photostynthetic rate in endolothic algae living under it is increased due to the large inflow of light, and excessive the photosynthetic products can be assimilated by animal cells [49].

Analysis of the published data and our own observations on damaged coral reefs allow us to think that algal colonization of a free coral reef substrate after catastrophes is a positive phenomenon rather than a negative one, which facilitates further restoration of the reef. First, by occupying free substrate, algae impede the attachment and growth of other more aggressive competitors on it, such as sponges (*Terpios hoshinota*), which are able to colonize not only the substrate but living corals as well [161]. Second, in most cases algae (the first settlers on a free substrate) are inferior to coral polyps in competitive ability, and within a period of not less than six months after a catastrophe they cannot be an obstacle for substrate colonization by corals [146]. Third, composition of algal communities is not permanent, and competitive algae can give way to noncompetitive ones. Fourth, algal turf communities, which are the most prevalent type on coral reefs, are not resistant to the effects of waves and from time to time in being torn off the substrate, the algal turf community clears it out, facilitating in such a way coral settlement on the reef. Fifth, in populating a substrate with algae, their growth and size and therefore their competitive abilities as well are controlled by herbivorous fish and invertebrates. Coral-eating fish and even invertebrates (except for the sea star *Acanthaster planci*) do not affect coral's competitiveness.

Transformation of coral reefs into vegetative reefs and their further destruction, which can be observed at the present time, are determined by two causes, a temporary increase in water temperature in shallow waters, which has been occurring frequently recently, and global or local anthropogenic pollution of the sea environment. There are some reasons to think that nature will find a way to rehabilitate coral reefs without human assistance, even under the conditions of repeated natural catastrophes. Restoration of coral reefs will probably take place due to the change of species composition of corals in shallow waters and their adaptation to high temperatures. However, under the complex effects of such phenomena as natural catastrophes and anthropogenic pollution of waters the coral reef ecosystem may be completely ruined within the next ten years.

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